

Total No. of Questions : 8]

SEAT No. :

PE-2479

[Total No. of Pages : 2

[6583]-3

T.E. (Civil)

**REMOTE SENSING & GEOGRAPHIC INFORMATION SYSTEM
(2019 Pattern) (Semester - VI) (301014)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume Suitable data, if necessary.*
- 4) *Use of electronic pocket calculator is allowed in the examination.*
- 5) *Neat diagrams must be drawn wherever necessary.*

- Q1)** a) Express the concept of augmentation and explain about satellite-based augmentation. **[6]**
- b) What are the three segments of GPS? Describe them briefly. **[6]**
- c) Illustrate the advantages and disadvantages of GNSS. **[5]**

OR

- Q2)** a) Describe briefly the concept of three-dimensional position location via intersection of multiple spheres. **[6]**
- b) Summarize the necessity of GNSS in various fields. **[6]**
- c) Describe the different sources of error in GPS. **[5]**

- Q3)** a) Differentiate between supervised and unsupervised image classification. **[6]**
- b) State the application of DEM. **[6]**
- c) Define Triangular Irregular Network Model (TIN) and its applications. **[6]**

OR

- Q4)** a) Write a note on digital image and analog image. **[6]**
- b) Explain in detail the concept of post processing and pre processing. **[6]**
- c) Differentiate between Visual and Digital image interpretation. **[6]**

P.T.O.

- Q5)** a) Illustrate Cloud computing with types and applications. [6]
b) Write in detail about various components of GIS. [6]
c) List down the differences between drafting software's and GIS. [5]

OR

- Q6)** a) Interpret the fundamentals of Cartography and map creation in GIS. [6]
b) Discuss the spatial and non-spatial data with example. [6]
c) Compare the various definitions of GIS used globally. [5]

- Q7)** a) Classify the Raster and Vector Data types. Explain with neat diagram. [6]
b) Differentiate various Attribute Data models and explain any one of them. [6]
c) Explain the application of GIS in Urban Planning. [6]

OR

- Q8)** a) Enlist the applications of GIS in Civil Engineering & Explain any one in detail. [6]
b) Outline the Georeferencing of GIS data? And explain its importance in the field of GIS? [6]
c) Categorize the different Database? Name the types of database. [6]

